

Supplementary

Complete Proofs and Complexity Analysis for **TriGAC**

A Theoretical Analysis

1) Unique-Cycle Property

Theorem A.1 (Unique-Cycle Property). *In the maximum co-cluster probability graph, each weakly connected component contains exactly one directed cycle.*

Proof. We prove the theorem by contradiction, structured in two parts.

Part 1: Existence of at least one directed cycle in a WCC.

Assume for contradiction that there exists a weakly connected component (WCC) C that contains no directed cycles. Let $n_C \triangleq |C|$ denote the number of objects in C . Since every object in C has exactly one outgoing edge to its maximum co-cluster probability neighbor, we can construct a directed path sequence starting from any object $x_i \in C$:

$$x_i \rightarrow \psi(x_i) \rightarrow \psi^2(x_i) \rightarrow \psi^3(x_i) \rightarrow \dots$$

where $\psi(x_i) \triangleq \arg \max_{x_k \neq x_i} p(x_i, x_k)$ denotes the maximum co-cluster probability neighbor of x_i , and $\psi^t(x_i)$ denotes the object reached by following t outgoing edges starting from x_i :

$$\psi^t(x_i) \triangleq \underbrace{\psi(\psi(\dots \psi(x_i) \dots))}_{t \text{ times}}.$$

Under the assumption that C contains no directed cycles, all objects in the sequence must be distinct.

(I) When $t = 1$, the untraversed object set is $C \setminus \{x_i, \psi(x_i)\}$, the number of traversed objects is 2, and the number of traversed edges is 1.

(II) When $t = n_C - 1$, the untraversed object set is $C \setminus \{x_i, \psi(x_i), \dots, \psi^{n_C-1}(x_i)\} = \emptyset$, the number of traversed objects is n_C , and the number of traversed edges is $n_C - 1$.

(III) When $t = n_C$, since $\psi^{n_C-1}(x_i) \in C$, it must have exactly one outgoing edge. However, the set of untraversed objects is empty. Therefore, $\psi^{n_C}(x_i)$ must be one of the objects already traversed. Let $\psi^{n_C}(x_i) = \psi^j(x_i)$ for some $0 \leq j \leq n_C - 1$. This forms a directed cycle:

$$\psi^j(x_i) \rightarrow \psi^{j+1}(x_i) \rightarrow \dots \rightarrow \psi^{n_C-1}(x_i) \rightarrow \psi^{n_C}(x_i) = \psi^j(x_i)$$

which contradicts the initial assumption that C contains no directed cycles. Therefore, every weakly connected component must contain at least one directed cycle.

Part 2: Uniqueness of the directed cycle in a WCC.

Assume for contradiction that there are two distinct directed cycles in C , denoted by $\mathcal{L}_1$ and $\mathcal{L}_2$. Let $V(\mathcal{L}_1)$ and $V(\mathcal{L}_2)$ be the sets of objects in cycles $\mathcal{L}_1$ and $\mathcal{L}_2$, respectively. We consider two cases based on whether these cycles share objects.

Case 1: $V(\mathcal{L}_1) \cap V(\mathcal{L}_2) = \emptyset$.

Since every object $x_i \in C$ has exactly one outgoing edge, the traversal sequence starting from any object is unique. As shown in Part 1, this path must eventually enter a directed cycle and remain within it. Consider any object $x \in V(\mathcal{L}_1)$, the unique path starting from x is entirely contained within $\mathcal{L}_1$, meaning $\psi^k(x) \in V(\mathcal{L}_1)$ for all $k \geq 0$. Similarly, for any object $y \in V(\mathcal{L}_2)$, the unique path starting from y is entirely contained within $\mathcal{L}_2$, meaning $\psi^k(y) \in V(\mathcal{L}_2)$ for all $k \geq 0$. Now, consider any object $z \in C$.

The path starting from z must eventually enter some directed cycle. If it enters $\mathcal{L}_1$, then all subsequent objects $\psi^k(z)$ (for sufficiently large k) belong to $V(\mathcal{L}_1)$. If it enters $\mathcal{L}_2$, they belong to $V(\mathcal{L}_2)$. Since $V(\mathcal{L}_1)$ and $V(\mathcal{L}_2)$ are disjoint, a path cannot lead to both cycles. Therefore, two node-disjoint directed cycles cannot exist in the same WCC.

Case 2: $\mathcal{L}_1$ and $\mathcal{L}_2$ share at least one object.

Let z be an object such that $z \in V(\mathcal{L}_1)$ and $z \in V(\mathcal{L}_2)$. Since z is in cycle $\mathcal{L}_1$, repeatedly applying the function ψ starting from z must trace the objects of $\mathcal{L}_1$ in sequence and eventually return to z . Let this sequence be:

$$z \rightarrow \psi(z) \rightarrow \cdots \rightarrow \psi^{l_1}(z) = z.$$

Similarly, since z is in cycle $\mathcal{L}_2$, repeatedly applying the function ψ starting from z must trace the objects of $\mathcal{L}_2$ in sequence and eventually return to z . Let this sequence be:

$$z \rightarrow \psi(z) \rightarrow \cdots \rightarrow \psi^{l_2}(z) = z.$$

Since each object has exactly one outgoing edge, the successor of z , which is $\psi(z)$, is unique. The successor of $\psi(z)$, which is $\psi^2(z)$, is also unique, and so on. Therefore, the sequence generated by starting at z and repeatedly applying ψ is entirely determined and unique. This unique sequence must form the directed cycle containing z . Since this sequence defines both $\mathcal{L}_1$ and $\mathcal{L}_2$, it must be that $\mathcal{L}_1$ and $\mathcal{L}_2$ are the exact same directed cycle. This contradicts the assumption that $\mathcal{L}_1$ and $\mathcal{L}_2$ were distinct directed cycles.

Both cases lead to a contradiction. Therefore, the assumption that a weakly connected component C contains more than one directed cycle is false. Combining the results from both parts, we conclude that every weakly connected component C contains exactly one directed cycle. $\square$

2) Cycle Compactness Property

Lemma A.2 (Distance Monotonicity). *Given a set of objects $\{x_1, \dots, x_n\}$, if $x_b = \psi(x_a)$ and $x_c = \psi(x_b)$, then*

$$d(x_a, x_b) \geq d(x_b, x_c). \quad (1)$$

Proof. Since $x_c = \psi(x_b)$ is the maximum co-cluster probability neighbor of x_b , and co-cluster probability decreases monotonically with distance, x_c is also the nearest neighbor of x_b in terms of distance. Therefore:

$$d(x_b, x_c) = \min_{x \neq x_b} d(x_b, x).$$

This implies that there does not exist any object x such that $d(x_b, x) < d(x_b, x_c)$, otherwise x_c would not be the maximum co-cluster probability neighbor of x_b . Consequently, we have:

$$d(x_a, x_b) \geq d(x_b, x_c).$$

$\square$

Corollary A.3 (Monotonicity along Outgoing Paths). *Given a WCC C in the maximum co-cluster probability graph, for any node $x_i \in C$, consider the path formed by repeatedly following outgoing edges:*

$$x_i \rightarrow \psi(x_i) \rightarrow \psi^2(x_i) \rightarrow \psi^3(x_i) \rightarrow \cdots$$

Then the outgoing edge lengths along this path are non-increasing:

$$d(x_i, \psi(x_i)) \geq d(\psi(x_i), \psi^2(x_i)) \geq d(\psi^2(x_i), \psi^3(x_i)) \geq \cdots \quad (2)$$

Proof. For any $\tau \geq 0$, consider two consecutive outgoing edges

$$\psi^\tau(x_i) \rightarrow \psi^{\tau+1}(x_i), \quad \psi^{\tau+1}(x_i) \rightarrow \psi^{\tau+2}(x_i).$$

Applying Lemma A.2 with $x_a = \psi^\tau(x_i)$, $x_b = \psi^{\tau+1}(x_i)$, and $x_c = \psi^{\tau+2}(x_i)$ yields

$$d(\psi^\tau(x_i), \psi^{\tau+1}(x_i)) \geq d(\psi^{\tau+1}(x_i), \psi^{\tau+2}(x_i)). \quad (3)$$

Chaining this inequality over $\tau = 0, 1, 2, \dots$ establishes the result. $\square$

Theorem A.4 (Cycle Compactness Property). *Let $\mathcal{L}$ denote a cycle in the maximum co-cluster probability graph, and let $\psi(\cdot)$ denote the outgoing neighbor with the maximum co-cluster probability. Then:*

(i) **Uniformity.** For any two cycle nodes $u, v \in \mathcal{L}$,

$$p(u, \psi(u)) = p(v, \psi(v)). \quad (4)$$

(ii) **Maximality.** For any cycle node $u \in \mathcal{L}$ and any off-cycle node $x \notin \mathcal{L}$,

$$p(u, \psi(u)) \geq p(x, \psi(x)). \quad (5)$$

Proof. Part (i). Consider any node $u_t \in \mathcal{L}$ and index the cycle nodes as

$$u_1 \rightarrow u_2 \rightarrow \cdots \rightarrow u_{n_{\mathcal{L}}} \rightarrow u_1,$$

where $u_{t+1} = \psi(u_t)$ for $t = 1, \dots, n_{\mathcal{L}}$ (with $u_{n_{\mathcal{L}}+1} = u_1$). Applying Corollary A.3 to the consecutive edges $u_t \rightarrow u_{t+1}$ and $u_{t+1} \rightarrow u_{t+2}$ yields

$$d(u_t, u_{t+1}) \geq d(u_{t+1}, u_{t+2}). \quad (6)$$

Chaining this inequality over all nodes on the cycle, we obtain the circular sequence

$$d(u_1, u_2) \geq \cdots \geq d(u_{n_{\mathcal{L}}-1}, u_{n_{\mathcal{L}}}) \geq d(u_{n_{\mathcal{L}}}, u_1) \geq d(u_1, u_2). \quad (7)$$

Since the leftmost and rightmost terms are identical, all intermediate inequalities must hold with equality. Therefore,

$$d(u_1, u_2) = \cdots = d(u_{n_{\mathcal{L}}}, u_1) \triangleq d_{\mathcal{L}}. \quad (8)$$

Because co-cluster probability decreases monotonically with distance, we have

$$p(u, \psi(u)) = p(v, \psi(v)), \quad \forall u, v \in \mathcal{L}. \quad (9)$$

Part (ii). For any off-cycle node $x \notin \mathcal{L}$, let its absorption path be

$$x \rightarrow \psi(x) \rightarrow \psi^2(x) \rightarrow \cdots \rightarrow \psi^{t-1}(x) \rightarrow u,$$

where $u = \psi^t(x) \in \mathcal{L}$ is the first node on the path that belongs to $\mathcal{L}$. By Corollary A.3, the edge lengths along this path are non-increasing:

$$d(x, \psi(x)) \geq \cdots \geq d(\psi^{t-1}(x), u) \geq d(u, \psi(u)). \quad (10)$$

Since $u \in \mathcal{L}$, Part (i) implies $d(u, \psi(u)) = d_{\mathcal{L}}$. Thus, by transitivity,

$$d(x, \psi(x)) \geq d_{\mathcal{L}}. \quad (11)$$

Again, because co-cluster probability decreases monotonically with distance, we have

$$p(u, \psi(u)) \geq p(x, \psi(x)), \quad \forall u \in \mathcal{L}, x \notin \mathcal{L}. \quad (12)$$

□

3) Expected Local-Density Plateau

Theorem A.4 establishes that the unique cycle $\mathcal{L}$ in each WCC is structurally compact: its outgoing edge lengths are uniform and minimal. We now show that, under a radial cluster structure, this structural compactness can be interpreted as a plateau of expected local density.

Definition A.5 (Local Density [1]). For a cutoff distance $d_c > 0$, the local density of object x_i within a WCC C is defined as

$$\rho_i(d_c) = \sum_{\substack{x_j \in C \\ j \neq i}} \mathbf{1}_{\{d(x_i, x_j) \leq d_c\}}. \quad (13)$$

Definition A.6 (Expected Local Density). Conditioned on the location of object x_i , the expected local density is

$$\mathbb{E}[\rho_i(d_c) \mid x_i]. \quad (14)$$

Assumption A.7 (Radial Cluster Structure). *The objects in each WCC C are drawn i.i.d. from a probability density $f(\cdot)$ that is continuous, bounded away from zero, and strictly decreasing along any radial direction from its unique mode μ_C .*

Lemma A.8 (Conditional Expectation of Local Density). *Conditioned on x_i , the expectation over the remaining $n_C - 1$ i.i.d. objects admits the closed form*

$$\mathbb{E}[\rho_i(d_c) \mid x_i] = (n_C - 1) \int_{B(x_i, d_c)} f(y) dy, \quad (15)$$

where $B(x_i, d_c)$ denotes the metric ball of radius d_c centered at x_i .

Proof. By linearity of expectation,

$$\mathbb{E}[\rho_i(d_c) \mid x_i] = \sum_{j \neq i} \mathbb{E}[\mathbf{1}_{\{d(x_i, x_j) \leq d_c\}} \mid x_i]. \quad (16)$$

For any $j \neq i$, the conditional expectation of the indicator equals the probability that x_j falls into $B(x_i, d_c)$:

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{\{d(x_i, x_j) \leq d_c\}} \mid x_i] &= \Pr(d(x_i, x_j) \leq d_c \mid x_i) \\ &= \int_{B(x_i, d_c)} f(y) dy. \end{aligned} \quad (17)$$

Summing over the $n_C - 1$ remaining i.i.d. objects yields the result. $\square$

Corollary A.9 (Expected Local-Density Plateau on the Cycle). *Let C be a weakly connected component (WCC) of the maximum co-cluster probability graph with unique directed cycle $\mathcal{L}$. Under Assumption A.7, for any cutoff $d_c > 0$, the cycle nodes form a plateau of expected local density:*

$$\mathbb{E}[\rho_u(d_c) \mid u] = \mathbb{E}[\rho_v(d_c) \mid v] \geq \mathbb{E}[\rho_x(d_c) \mid x], \quad (18)$$

for all $u, v \in \mathcal{L}$ and $x \notin \mathcal{L}$.

Proof. By Theorem A.4, the outgoing edge lengths of cycle nodes are uniform and minimal:

$$d(u, \psi(u)) = d(v, \psi(v)) = d_{\mathcal{L}} \leq d(x, \psi(x)), \quad (19)$$

for all $u, v \in \mathcal{L}$ and $x \notin \mathcal{L}$.

Under Assumption A.7, the density admits the form $f(y) = g(\|y - \mu_C\|)$ with $g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ strictly decreasing. Consequently, the nearest-neighbor distance $R_y = d(y, \psi(y))$ serves as an inverse proxy for the radial distance $\|y - \mu_C\|$: a smaller R_y indicates that y lies in a more concentrated region closer to the mode μ_C , and thus $f(y)$ is higher. Therefore, the uniform and minimal edge lengths on the cycle imply that all cycle nodes are at the same radial distance from μ_C and closer than any off-cycle node. Formally,

$$\|u - \mu_C\| = \|v - \mu_C\| \leq \|x - \mu_C\|, \quad (20)$$

for all $u, v \in \mathcal{L}$ and $x \notin \mathcal{L}$.

Because $f(\cdot)$ is strictly decreasing along radial directions, the average density within any metric ball of fixed radius d_c is higher when the ball is centered closer to μ_C . Hence,

$$\int_{B(u, d_c)} f(y) dy = \int_{B(v, d_c)} f(y) dy \geq \int_{B(x, d_c)} f(y) dy. \quad (21)$$

Multiplying by the positive factor $(n_C - 1)$ and applying Lemma A.8, we obtain

$$\mathbb{E}[\rho_u(d_c) \mid u] = \mathbb{E}[\rho_v(d_c) \mid v] \geq \mathbb{E}[\rho_x(d_c) \mid x], \quad (22)$$

which completes the proof. $\square$

4) Upper Bound of Pairwise Constraints

Theorem A.10. *Given a multi-view dataset $\mathcal{X} = \{X^{(v)}\}_{v=1}^m$ with n samples and k ground-truth clusters, **TriGAC** requires at most $nk - \frac{k^2}{2} - \frac{k}{2}$ pairwise constraints to achieve the optimal clustering partition.*

Proof. Consider the worst-case scenario where the first k queried nodes each belong to distinct ground-truth clusters. Initially, the constraint graph $\mathcal{G}_{\text{cons},0}$ is empty, so the first queried node initializes the first WCC with zero queries.

For the i -th queried node ($2 \leq i \leq k$), there exist $(i-1)$ WCCs in $\mathcal{G}_{\text{cons}}$. In the worst case, this node receives a cannot-link (CL) verdict against all existing WCCs, so the algorithm exhausts the entire annotation sequence of length $(i-1)$ and then creates a new singleton WCC. The number of queries consumed is $(i-1)$. Summing over $i = 2, \dots, k$ yields $\sum_{i=2}^k (i-1) = \frac{k(k-1)}{2}$.

For each remaining node ($k+1 \leq i \leq n$), there are exactly k WCCs. Since there are only k ground-truth clusters, this node must share a must-link (ML) with one of them. In the worst case, it receives a CL against the first $(k-1)$ WCCs and shares an ML only with the last one, consuming k queries. Summing over $i = k+1, \dots, n$ yields $k(n-k)$.

Therefore, the total number of queries in the worst case is

$$\frac{k(k-1)}{2} + k(n-k) = nk - \frac{k^2}{2} - \frac{k}{2}.$$

□

B Complexity Analysis

TriGAC adopts the Euclidean distance as the default metric. However, Euclidean distance suffers from the curse of dimensionality on high-dimensional features [2]. To handle high-dimensional data without incurring scalability penalties, we apply a lightweight two-stage projection: randomized PCA [3] reduces each view from the original dimension D_v to 50 dimensions in $\mathcal{O}(nD_v)$ time, followed by UMAP [4] to 20 dimensions via approximate nearest neighbors in $\mathcal{O}(n \log n)$ time. Since D_v , 50, and 20 are constants independent of the sample size n , the overall preprocessing cost is $\mathcal{O}(n \log n)$ and is absorbed by the subsequent $\mathcal{O}(n \log n)$ steps. Consequently, the asymptotic time complexity derived below remains intact.

We now analyze the time cost of each step and summarize the overall complexity.

KNN local region construction. For each view, computing the KNN local regions over n objects can be accomplished in $\mathcal{O}(n \log n)$ time by utilizing the ball-tree technique [5]. With m views, the time complexity is $\mathcal{F}_1 \in \mathcal{O}(mn \log n)$.

Object graph initialization. In each view, the iterative agent-object merging repeatedly queries the maximum co-cluster probability neighbor for every object in the current agent set A . Utilizing the ball-tree, each search costs $\mathcal{O}(\log n)$, and each iteration processes $|A|$ objects. Since $|A|$ is at least halved in every iteration (in the worst case each weakly connected component contains only two objects), the total time per view is bounded by

$$\mathcal{O}\left(n \log n + \frac{n}{2} \log \frac{n}{2} + \dots\right) = \mathcal{O}(n \log n). \quad (23)$$

Thus, with m views, the time complexity is $\mathcal{F}_2 \in \mathcal{O}(mn \log n)$.

Representativeness and uncertainty scores calculation. Computing the representativeness score $\mathcal{H}^{(v)}$ requires a single pass over the n objects in each view, giving $\mathcal{O}(mn)$. For the uncertainty score $\mathcal{U}_t^{(v)}$, the KNN regions are already available; computing the local entropy per object is $\mathcal{O}(K) = \mathcal{O}(1)$. Over T rounds, the time complexity is $\mathcal{F}_3 = \mathcal{O}(Tmn)$.

Informative object selection. In each round, the top- b_t candidates are extracted from m per-view pools. By maintaining a global max-heap for all n objects across views, each extraction and score update costs $\mathcal{O}(\log n)$. Since every object is extracted at most once during the T rounds, the time complexity is $\mathcal{F}_4 = \mathcal{O}(mn \log n)$.

Constraint graph update. For each candidate object x_i , we inspect at most M WCCs in the current constraint graph. Because the graph only grows by merging objects via must-link constraints, M is bounded by the ground-truth cluster number k . For each WCC, we scan at most Ω representative objects across m views to compute the fused distance, which costs $\mathcal{O}(m\Omega)$. Sorting the M anchors takes $\mathcal{O}(M \log M)$. In the worst case, x_i is queried against all M WCCs before a must-link is found or a

new singleton WCC is created, adding $\mathcal{O}(M)$. Hence, processing one object costs $\mathcal{O}(k(m\Omega + \log k))$. In the worst case, **TriGAC** processes at most n objects over all T rounds, so the total cost is $\mathcal{F}_5 = \mathcal{O}(nk(m\Omega + \log k))$.

Object graph refinement and consensus fusion. The refinement and consensus fusion scans all m views for every object, costing $\mathcal{O}(mn)$ per round. Over T rounds, the total refinement cost is $\mathcal{F}_6 = \mathcal{O}(Tmn)$.

View graph construction and view confidence update. Computing NMI between all pairs of m per-view partitions takes $\mathcal{O}(m^2n)$ per round because NMI is linear in n for a given partition. Building the view graph and updating the confidence weights over m views costs $\mathcal{O}(m^2)$. Over T rounds, this yields $\mathcal{F}_7 = \mathcal{O}(Tm^2n)$.

Summary. Summing up all steps, in the worst case (when all the objects are selected), the overall time complexity of **TriGAC** is

$$\mathcal{F} = \sum_{i=1}^7 \mathcal{F}_i \in \mathcal{O}(Tm^2n + mn \log n + nk(m\Omega + \log k)). \quad (24)$$

When m , T , k , and Ω are treated as small constants, **TriGAC** scales quasi-linearly with the number of samples, i.e., $\mathcal{O}(n \log n)$.

Space Complexity. **TriGAC** mainly stores the sparse graph structures, KNN local regions, representative windows, and auxiliary scores. Since each node in an object graph has at most one outgoing edge, the object graphs, constraint graph, and view graph require only linear storage with respect to the number of involved nodes. Together with the KNN local regions and auxiliary scores, this gives $\mathcal{O}(mn)$ space. The additional representative windows require $\mathcal{O}(mM\Omega)$ space, where M denotes the number of constraint-graph WCCs and Ω is the window size. Since the constraint graph never contains more than k WCCs ($M \leq k$), the worst-case storage is $\mathcal{O}(mk\Omega)$. Excluding raw feature storage, the overall space complexity in the worst case is

$$\mathcal{S} \in \mathcal{O}(mn + mk\Omega), \quad (25)$$

which reduces to $\mathcal{O}(mn)$ when M and Ω are treated as constants.

References

- [1] A. Rodriguez and A. Laio, “Clustering by fast search and find of density peaks,” *science*, vol. 344, no. 6191, pp. 1492–1496, 2014.
- [2] C. C. Aggarwal, A. Hinneburg, and D. A. Keim, “On the surprising behavior of distance metrics in high dimensional space,” in *International conference on database theory*. Springer, 2001, pp. 420–434.
- [3] N. Halko, P.-G. Martinsson, and J. A. Tropp, “Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions,” *SIAM review*, vol. 53, no. 2, pp. 217–288, 2011.
- [4] J. Healy and L. McInnes, “Uniform manifold approximation and projection,” *Nature Reviews Methods Primers*, vol. 4, no. 1, p. 82, 2024.
- [5] Q. Huang and A. K. Tung, “Lightweight-yet-efficient: Revitalizing ball-tree for point-to-hyperplane nearest neighbor search,” in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*. Anaheim, California, USA: IEEE, 2023, pp. 436–449.